

```
In [19]: # Exp_6
# Name:Suhani Milind Gayakwad
# Roll no:16
# Sec:A
# Subject:ET1
# Date:8/09/2025
```

```
In [20]: #importing the basic Library
import numpy as np
from matplotlib import pyplot as plt
```

```
In [21]: x=np.arange(1,11)
```

```
In [22]: x
```

```
Out[22]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

```
In [23]: print(x)

[ 1  2  3  4  5  6  7  8  9 10]
```

```
In [24]: y=2*x
```

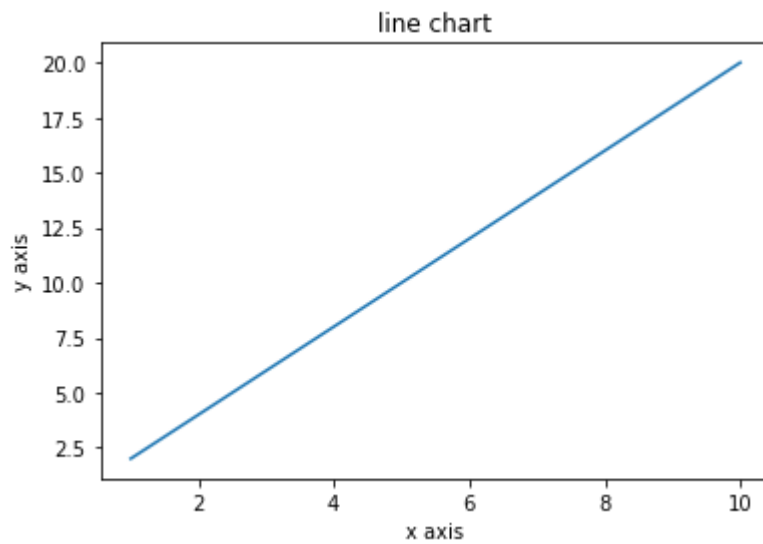
```
In [25]: y
```

```
Out[25]: array([ 2,  4,  6,  8, 10, 12, 14, 16, 18, 20])
```

line chart

```
In [26]: plt.plot(x,y)
plt.title("line chart")
plt.xlabel("x axis")

plt.ylabel("y axis")
plt.show()
```

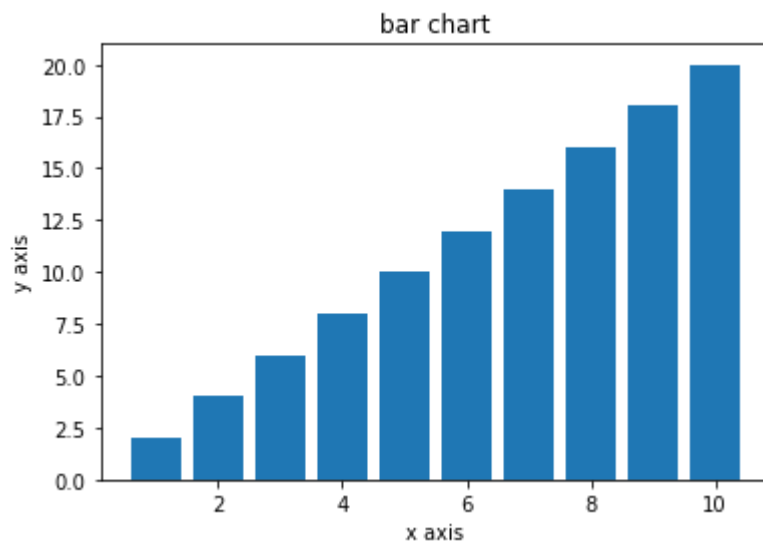


BARCHART

In [27]:

```
plt.bar(x,y)
plt.title("bar chart")
plt.xlabel("x axis")

plt.ylabel("y axis")
plt.show()
```



In [29]:

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate random numbers
# For example, 10 random integers between 1 and 50
data = np.random.randint(1, 50, size=10)

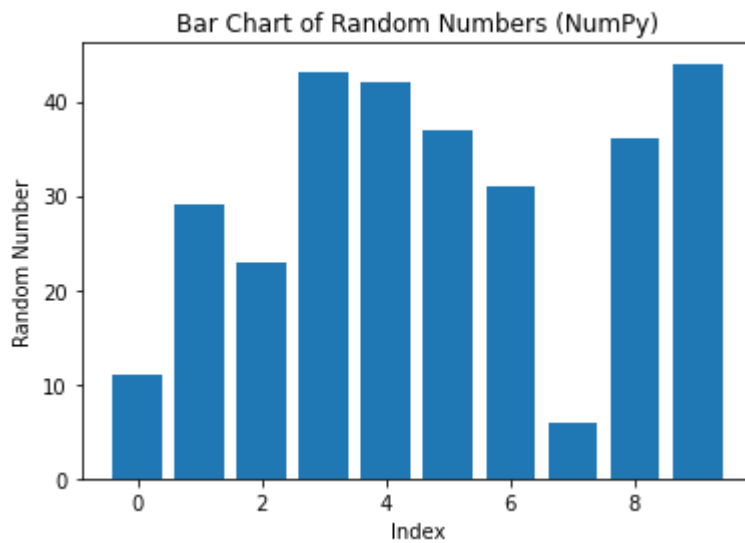
# Step 2: Create x-axis values (like labels for each bar)
x = np.arange(len(data))

# Step 3: Plot bar chart
plt.bar(x, data)

# Step 4: Add labels and title
plt.xlabel("Index")
plt.ylabel("Random Number")
```

```
plt.title("Bar Chart of Random Numbers (NumPy)")

# Step 5: Show chart
plt.show()
```



In [43]:

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate random numbers
data = np.random.randint(1, 20, size=8) # 8 random numbers between 1 and 20

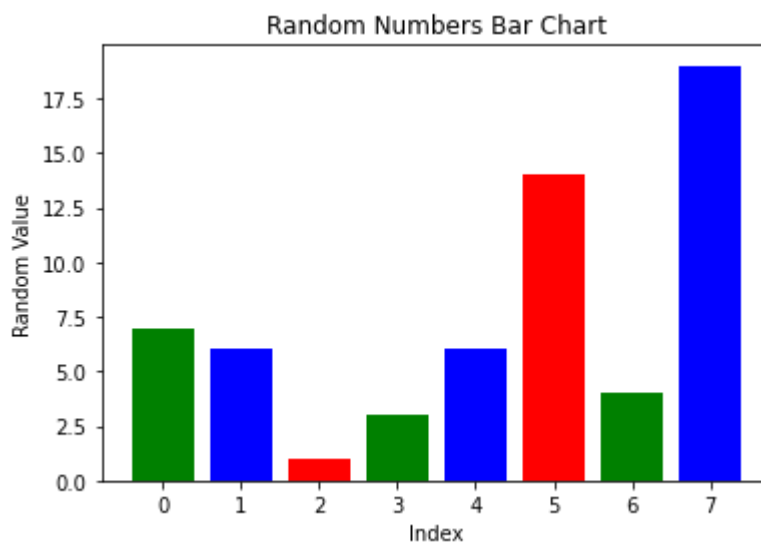
# Step 2: Create bar chart
x = np.arange(len(data)) # positions for bars

# Step 3: Change bar colors
colors = ['green', 'blue', 'red']

plt.bar(x, data, color=colors)

# Add Labels
plt.xlabel("Index")
plt.ylabel("Random Value")
plt.title("Random Numbers Bar Chart")

# Show plot
plt.show()
```



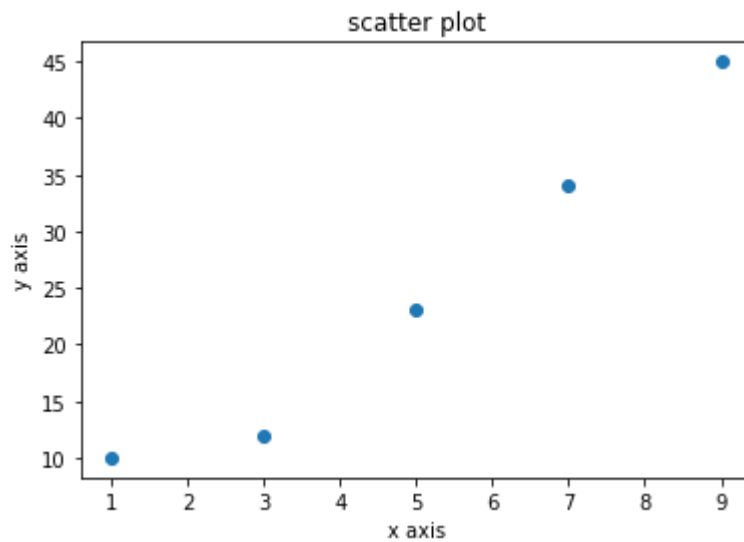
scatter plot

In [34]:

```
a=(1,5,9,3,7)
b=(10,23,45,12,34)

plt.scatter(a,b)
plt.title("scatter plot")
plt.xlabel("x axis")

plt.ylabel("y axis")
plt.show()
```



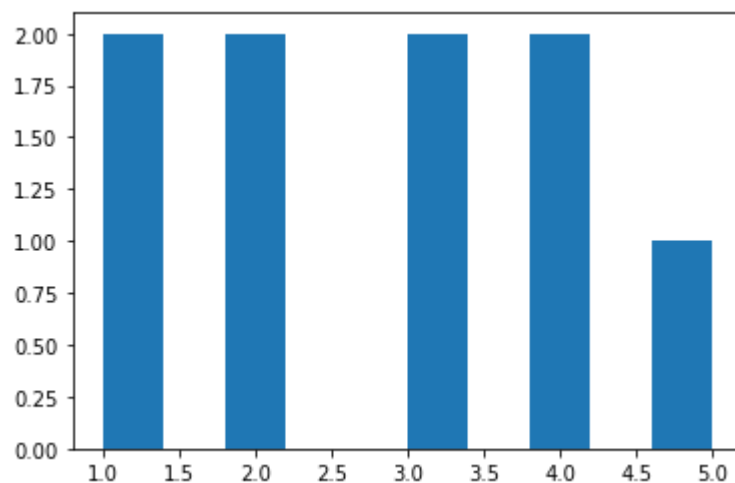
HISTRO-GRAM

In [44]:

```
h=(1,2,3,4,5,1,2,3,4,8,9,9,9)
```

In [46]:

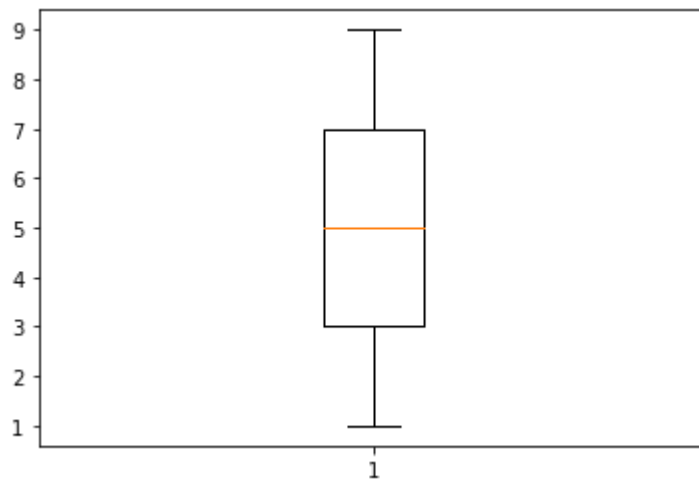
```
plt.hist(H)
plt.show()
```



BOXPLOT

```
In [48]: B=[1,2,3,4,5,6,7,8,9]
```

```
In [50]: plt.boxplot(B)  
plt.show()
```



```
In [ ]:
```